BASICS OF COMPUTER NETWORKS · COURSE 04

Day 05 — Application Services & Wi-Fi

Client–Server Interaction · Wi-Fi Technology · Lab Build

Cohort · Cyber.SoHo Networking Essentials (NetEss 3.0 aligned)

Session 5 of 6 · Tuesday, May 12, 2026

4 contact hours (2 h lectures + 2 h labs) + ~2 h home practice

From the Industry to the Classroom — Building the IT's Future, Today and Together!

Agenda — Today's Mission

Turn the wired neighborhood built on Day 04 into a city — names, web service, wireless clients.

1 **Lecture 1** — 60 min

Client–Server Interaction

2 **Exercise 1** — 60 min

Wireshark application-services lab

3 **Lecture 2** — 60 min

Wi-Fi Technology

4 **Exercise 2** — 60 min

Project Stage 5 — Services + Wireless

KEY TAKEAWAY

By end of today, Acme SoHo will resolve names, serve a web page, and accept Wi-Fi clients.

LECTURE 1 · 60 MIN

Client–Server Interaction

Transport layer, ports, VLANs, application services.

One conversation at a time — and how the OS routes it.

1.1 — TCP vs UDP at the Transport Layer (1/3)

Two completely different conversation styles at Layer 4

What lives at Layer 4 — the Transport Layer

- TCP — Transmission Control Protocol → reliable, ordered, connection-oriented
- UDP — User Datagram Protocol → connectionless, unordered, best-effort

TCP — the protocol that calls you back

- Three-way handshake before any data:
  - Client → Server: SYN (synchronize)
  - Server → Client: SYN-ACK
  - Client → Server: ACK (let's go)
- Every byte numbered → retransmits losses, reorders out-of-order segments
- Closes cleanly with FIN
- Cost: 20-byte minimum header + per-connection bookkeeping

1.1 — TCP vs UDP at the Transport Layer (2/3)

UDP — and why "unreliable" is sometimes the right answer

UDP — the protocol that just shouts and walks away

- No handshake, no ACKs, no retransmits
- 8-byte header (TCP's is 20+)
- Application does its own reliability — if it wants any
- Lower latency, lower overhead

Why two protocols exist

Not every message wants the same guarantees.

- For some traffic, the application can do reliability cheaper than TCP can
- For other traffic, a missed packet is worse than a late packet
- VoIP, live video, DNS lookups, online games → all prefer UDP

1.1 — TCP vs UDP at the Transport Layer (3/3)

When to pick which

Use TCP for…

- Web (HTTP / HTTPS)
- Email submission & retrieval (SMTP, IMAP)
- SSH and file transfer (SFTP, FTPS)
- Anywhere bytes must all arrive, in order
- Anywhere a connection state actually helps

Use UDP for…

- DNS lookups (UDP/53)
- DHCP address negotiation
- VoIP, live audio & video calls
- Online games — low-latency packet streams
- Anywhere a late packet is worse than a lost one

KEY TAKEAWAY

TCP = reliability with overhead. UDP = speed with no promises. The application chooses.

1.2 — Well-Known Port Numbers (1/3)

What a port is and how the OS uses it

What is a port?

- A 16-bit number (0 – 65,535) that tells the OS which application a packet belongs to
- The pair {IP address + port} is called a socket

Three ranges (IANA)

- 0 – 1023 · Well-known / System · standard server-side services (root/admin to bind on Linux)
- 1024 – 49,151 · Registered · reserved for specific vendors / apps
- 49,152 – 65,535 · Dynamic / Ephemeral · client-side, OS-assigned per outbound connection

Mental model

- Ports < 1024 = the post office's named windows
- Ports 49,152+ = the customer's return address — different each visit

1.2 — Well-Known Port Numbers (2/3)

The top 10 every networker memorises

PORT	PROTOCOL	TRANSPORT	WHAT IT DOES
20 / 21	FTP	TCP	File transfer (data / control)
22	SSH	TCP	Secure remote shell + file copy
23	Telnet	TCP	Plain-text remote shell (legacy)
25	SMTP	TCP	Mail submission / relay
53	DNS	UDP + TCP	Name resolution
67 / 68	DHCP	UDP	Server / client lease negotiation
80	HTTP	TCP	Web — plaintext
110	POP3	TCP	Mail retrieval (download-and-delete)
143	IMAP	TCP	Mail retrieval (server-side folders)
443	HTTPS	TCP	Encrypted web (TLS)

KEY TAKEAWAY

Memorise these 10 ports. They are the street signs of the internet — without them you can't read a capture.

1.2 — Well-Known Port Numbers (3/3)

Sockets — how the OS routes packets to the right program

The socket 4-tuple

Every TCP connection on every machine is uniquely identified by the tuple:

```
{ source IP : source port  ↔  destination IP : destination port }
```

What this means in practice

- One server at 192.168.1.10 can run a web service on port 80 AND an SSH service on port 22 simultaneously
- Your browser opens 6 simultaneous TCP connections to the same web server — each uses a different source port
- When the OS receives a packet, it looks up the 4-tuple in its socket table to find the owning process

Quick OS commands to inspect ports

```
Linux:    ss -tulpn
macOS:    lsof -iTCP -sTCP:LISTEN -P -n
Windows:  netstat -an  |  findstr LISTENING
```

1.3 — LAN Segmentation & VLANs (1/3)

Why one flat LAN stops scaling

The broadcast-domain problem

- One flat LAN = one broadcast domain
- Every ARP request, every DHCP DISCOVER, every Wake-on-LAN frame reaches every device
- 50 hosts → manageable · 500 hosts → broadcast storms and security incidents

Three reasons to segment

- Performance — contain broadcast traffic
- Security — isolate sensitive groups (servers, finance, guest Wi-Fi)
- Manageability — apply QoS / ACL / DHCP scopes per group

1.3 — LAN Segmentation & VLANs (2/3)

IEEE 802.1Q — tag-based logical segmentation

Virtual LANs

- One physical switch behaves like multiple logical switches
- Frames are tagged with a 12-bit VLAN ID (1 – 4094)

Port roles

- Access port — carries ONE VLAN, tag stripped before the host sees it
- Trunk port — carries MULTIPLE VLANs between switches and routers

Moving between VLANs

- Two devices on different VLANs cannot talk at Layer 2 — even on the same switch
- A Layer-3 device must route between them (router-on-a-stick or L3 switch)

1.3 — LAN Segmentation & VLANs (3/3)

Acme SoHo VLAN plan — used in Exercise 2

VLAN	NAME	SUBNET	PURPOSE
10	STAFF	192.168.10.0/24	8 employee endpoints
20	SERVERS	192.168.20.0/24	Internal DNS, HTTP, file
30	VOICE	192.168.30.0/24	VoIP phones
40	GUEST	192.168.40.0/24	Visitor Wi-Fi (internet-only)
99	MGMT	192.168.99.0/24	Switch / AP admin

KEY TAKEAWAY

VLANs slice one physical switch into many logical switches — better performance, better security, no extra hardware.

1.4 — Application-Layer Services (ARP — 1 of 6)

Address Resolution Protocol

A. ARP

`Layer 2/3 boundary · no port (Ethertype 0x0806)`

- Maps a Layer-3 IPv4 address → Layer-2 MAC address on the same broadcast domain
- Step 1: host broadcasts "Who has 192.168.1.1? Tell 192.168.1.50."
- Step 2: owner replies UNICAST with its MAC
- Step 3: both sides cache the binding (arp -a)
- Critical weakness: NO authentication → ARP-spoofing is the classic LAN Man-in-the-Middle technique
- Mitigations: Dynamic ARP Inspection (DAI), port security, static ARP entries

1.4 — Application-Layer Services (DNS — 2 of 6)

Domain Name System

B. DNS

`UDP/53 default · TCP/53 for large responses · DoH 443 · DoT 853`

- Translates human-friendly names → IP addresses (and back, mail records, service records, etc.)
- Hierarchical lookup: root "." → TLD ".com" → authoritative "example.com"
- Record types: A (IPv4), AAAA (IPv6), CNAME (alias), MX (mail), TXT (anything)
- Default transport UDP/53 (fast); TCP/53 for large responses and zone transfers
- Modern variants: DoH (DNS-over-HTTPS), DoT (DNS-over-TLS) — encrypted lookups

1.4 — Application-Layer Services (HTTP — 3 of 6)

Hypertext Transfer Protocol / HTML

C. HTTP / HTML

`HTTP TCP/80 · HTTPS TCP/443 · HTTP/3 over QUIC UDP/443`

- Request / response over TCP — the protocol of the Web
- Request line example: GET /index.html HTTP/1.1
- Headers describe browser, language, cookies, content type
- Status codes: 200 OK · 301 Moved · 404 Not Found · 500 Server Error
- HTTPS = HTTP wrapped in TLS — same protocol, encrypted
- HTML (Hypertext Markup Language) is the document the server returns

1.4 — Application-Layer Services (FTP — 4 of 6)

File Transfer Protocol

D. FTP

`Control TCP/21 · Data TCP/20 (active) or random (passive)`

- Two channels: control TCP/21 + data TCP/20 (active) or random ephemeral (passive)
- Authentication and commands are PLAINTEXT — never run over an untrusted network
- Use SFTP instead — SSH File Transfer Protocol, runs inside SSH/22
- Or FTPS — FTP wrapped in TLS
- Modern alternatives: SCP, rsync over SSH, HTTPS-based object storage

1.4 — Application-Layer Services (Telnet vs SSH — 5 of 6)

Remote shell — legacy vs current

E. Telnet vs SSH

`Telnet TCP/23 · SSH TCP/22`

- Telnet: plaintext, no encryption, credentials in clear — LEGACY, avoid in production
- SSH: strong encryption (AES, ChaCha20), password OR public-key authentication
- SSH adds: port forwarding, file copy (scp / sftp), X11 forwarding, tunneling
- Rule: never run Telnet across a network you do not physically control
- Disable Telnet on every switch and router during initial hardening

1.4 — Application-Layer Services (Email — 6 of 6)

Mail protocols: SMTP · POP3 · IMAP

F. Email Protocols

`SMTP 25 / 587 / 465 · POP3 110 / 995 · IMAP 143 / 993`

- SMTP — sends and relays mail between servers (port 25) and from clients (587 STARTTLS, 465 implicit TLS)
- POP3 — downloads mail to one device, then deletes from server — good for one device, bad for multi-device
- IMAP — keeps mail on the server, multi-device sync — what Gmail, Outlook, and Apple Mail all use
- All three have TLS variants (465 / 995 / 993) — prefer those over the plaintext ports

KEY TAKEAWAY

ARP finds neighbours, DNS finds names, HTTP serves pages, FTP/SSH move files, SMTP/POP3/IMAP move mail.

EXERCISE 1 · 60 MIN IN CLASS + 60 MIN HOME

Application Services Lab

Capture and decode ARP, DNS, and HTTP in Wireshark.

Optional but recommended.

Exercise 1 — Wireshark Captures (1/2)

60 min in class + 60 min home · Optional but recommended

Lab goal

Capture and decode three real packets in Wireshark: ARP, DNS, and HTTP GET.

Setup (10 min)

- VirtualBox VM or host with Wireshark installed
- Pick a live interface (Wi-Fi or loopback)
- Apply display filter from a clean state

Part A — ARP (15 min)

- Clear ARP cache: `ip neigh flush all` (Linux) / `arp -d *` (Windows, admin)
- Filter: arp
- Ping the default gateway → verify broadcast "Who has X?" then unicast "X is at MAC"

Part B — DNS (15 min)

- Filter: dns · run nslookup or dig
- Identify Query (A record), Response, TTL, server IP
- Bonus: capture UDP/53 and one TCP/53 (large response)

Exercise 1 — Wireshark Captures (2/2)

Part C, deliverable, and alternative tools

Part C — HTTP GET (20 min)

- Filter: http.request.method == "GET"
- Browse to a plain-HTTP test site or local lab server
- Right-click the GET packet → Follow → HTTP Stream
- Read: request line, host header, user-agent, status code, content-type

Deliverable

- 1-page lab note with three screenshots (one per protocol) and one sentence each on what you observed

Alternative open-source tools

- tshark — Wireshark's CLI sibling
- tcpdump — the classic UNIX packet sniffer
- Ettercap / Bettercap — interactive network analyzers
- netcat / ncat — manual TCP / UDP probes
- mitmproxy — interactive HTTPS inspection (own lab only)

LECTURE 2 · 60 MIN

Wi-Fi Technology

From the box on the shelf to the protocol on the wire.

Standards, channels, security.

2.1 — The Integrated Wireless Router (1/2)

One plastic case · four logical devices

What the box on your shelf actually contains

- Router — Layer-3 forwarding plus NAT (Network Address Translation)
- Switch — Layer-2 wired ports, usually 4 × Gigabit
- Wireless AP (Access Point) — 802.11 radio(s)
- Stateful firewall — blocks unsolicited inbound traffic

Plus: DHCP server, DNS proxy, web admin UI, sometimes a USB share

Why this matters

- One device = simple for home use
- Enterprise reality: each role is a separate appliance (better performance, manageability)
- The same logical functions exist either way

2.1 — The Integrated Wireless Router (2/2)

Acme SoHo's box — naming each port and radio

WAN port

- Fibre / cable / 4G uplink to the ISP

4 LAN ports

- Wired workstations — patch cables to the office desks

2.4 GHz + 5 GHz radios

- Staff laptops, phones, guest Wi-Fi
- Separate SSIDs per band where possible (range vs speed)

Admin UI

- `Reachable at 192.168.1.1 via HTTP or HTTPS`
- Change the default password during initial setup

KEY TAKEAWAY

A home router is four devices in one box. Knowing which sub-device handles your problem is half the troubleshooting.

2.2 — Wired vs Wireless (1/2)

Side-by-side comparison

PROPERTY	WIRED (Ethernet)	WIRELESS (Wi-Fi)
Medium	Copper twisted-pair / fibre	Radio (2.4 / 5 / 6 GHz)
Topology	Star (each host → switch)	Shared cell (every host → AP)
Bandwidth	1 / 2.5 / 10 / 40 / 100 Gbps	600 Mbps – 9.6 Gbps (shared)
Latency	< 1 ms typical	2 – 10 ms typical, can spike
Jitter	Very low	Variable (interference)
Security	Physical access required	Anyone in range can listen
Reliability	Predictable	Depends on RF environment
Mobility	None	Full

2.2 — Wired vs Wireless (2/2)

When to pick which · how media access works

When to pick which

- Wired: servers, VoIP, video editing, gaming — anywhere predictability matters
- Wireless: laptops, phones, tablets, IoT, guests — anywhere mobility matters

Media access — CSMA

- Wired Ethernet uses CSMA/CD (Collision Detection)
  - Full-duplex on modern switches makes collisions impossible
- Wi-Fi uses CSMA/CA (Collision Avoidance) — covered in §2.4
  - A radio is one-or-the-other: cannot listen and talk at the same time

KEY TAKEAWAY

Wire what stays put. Radio what moves. Mix them, document them, and stop apologising for the cables.

2.3 — Wi-Fi Standards (1/2)

The IEEE 802.11 family tree

STANDARD	MARKETING	YEAR	BAND(S)	MAX RATE
802.11a	—	1999	5 GHz	54 Mbps
802.11b	—	1999	2.4 GHz	11 Mbps
802.11g	—	2003	2.4 GHz	54 Mbps
802.11n	Wi-Fi 4	2009	2.4 + 5 GHz	600 Mbps
802.11ac	Wi-Fi 5	2013	5 GHz	6.9 Gbps
802.11ax	Wi-Fi 6/6E	2019 / 21	2.4 + 5 (+6)	9.6 Gbps
802.11be	Wi-Fi 7	2024	2.4 + 5 + 6	46 Gbps

KEY TAKEAWAY

Higher Wi-Fi generation = more antennas, more modulation, cleaner channels. Backward compatibility slows everyone to the oldest client.

2.3 — Wi-Fi Standards (2/2)

Key acronyms and the three bands

Key terms

- MIMO — Multiple Input Multiple Output — multiple antennas on each side
- MU-MIMO — Multi-User MIMO — talk to several clients simultaneously
- OFDMA — Orthogonal Frequency Division Multiple Access — sub-divide a channel into resource units (Wi-Fi 6)
- QAM — Quadrature Amplitude Modulation — more bits per symbol = faster, but needs cleaner signal
- MLO — Multi-Link Operation — one client uses 2 or 3 bands in parallel (Wi-Fi 7)

Bands — rules of thumb

- 2.4 GHz — longer range, slower, crowded (Bluetooth, microwaves, baby monitors)
- 5 GHz — shorter range, faster, cleaner
- 6 GHz — brand-new, very clean — Wi-Fi 6E / 7 only

2.4 — How Wireless Traffic is Controlled (1/2)

CSMA/CA and channel planning

CSMA/CA — Carrier Sense Multiple Access with Collision Avoidance

- Wi-Fi clients cannot detect collisions WHILE transmitting (one radio, can't listen and talk)
- So they avoid collisions instead. The protocol:
  1. Listen — is the channel free?
  2. If busy → wait until quiet
  3. Wait an additional random back-off time
  4. Transmit
  5. Wait for ACK from receiver · no ACK → assume collision → back off and retry

Channels

- 2.4 GHz: channels 1 – 13 (EU) · only 1, 6, 11 are non-overlapping
- 5 GHz: 23+ channels, mostly non-overlapping · DFS rules apply
- Co-channel interference = neighbours on the same channel wait their turn

2.4 — How Wireless Traffic is Controlled (2/2)

SSID, beacons, and association

SSID — Service Set Identifier

- Human-readable network name, up to 32 bytes
- Broadcast in beacons by the AP (~10/sec, default 100 ms TBTT)
- Hidden SSID is NOT security — clients still leak it in probe requests
- BSSID = the AP's MAC address (unique per radio)
- ESS = multiple APs sharing the same SSID for roaming

Beacons & association sequence

1. AP broadcasts beacons advertising SSID, supported rates, capabilities
2. Client sends probe request, AP replies with probe response
3. Authentication (open or WPA-based)
4. Association — client is now a member of the BSS

KEY TAKEAWAY

Radio is shared. CSMA/CA is the polite protocol that lets many clients share one channel.

2.5 — Secure Wi-Fi: WPA3 / PSK (1/2)

Security history — and why WEP and WPA2 are not enough

STANDARD	YEAR	STATUS
WEP — Wired Equivalent Privacy	1997	Broken — never use
WPA — Wi-Fi Protected Access	2003	Stopgap
WPA2 (802.11i, AES-CCMP)	2004	Industry default 2006 – 2020
WPA3	2018	Current best practice
WPA3-SAE (Personal)	2018	Replaces WPA2-PSK

2.5 — Secure Wi-Fi: WPA3 / PSK (2/2)

What WPA3 fixes · PSK basics · safer defaults

What WPA3 fixes

- SAE — Simultaneous Authentication of Equals — replaces the 4-way handshake → kills offline dictionary attack
- Forward secrecy — capturing today's traffic does not decrypt yesterday's
- PMF — Protected Management Frames — stops deauth attacks
- 192-bit Enterprise mode for high-security environments

PSK — Pre-Shared Key

- A passphrase (8 – 63 chars) shared by AP and every client
- AP and client derive a per-session key from the passphrase
- Length beats complexity — 16+ random chars is effectively unbreakable today

Safer defaults

- Separate SSID for guests (internet-only)
- Disable WPS (Wi-Fi Protected Setup)
- Keep firmware updated · prefer WPA3-Personal when all clients support it

EXERCISE 2 · STAGE 5 OF THE BUILD CHALLENGE

Services + Wireless

DNS + HTTP server on the SERVERS VLAN — optional Wi-Fi segment.

Reach the intranet by name, not by IP.

Exercise 2 — Stage 5: Services + Wireless (1/2)

Required: stand up DNS + HTTP on the SERVERS VLAN

Stage 5 goals

- Add a DNS server to the SERVERS VLAN — resolves *.acme.lan
- Add an HTTP server that serves a one-page intranet
- Verify a STAFF VLAN client reaches http://www.acme.lan by name
- Optional but recommended: add a Wi-Fi access segment

Required steps (60 min)

1. Step 1 — In GNS3, attach a Linux VM to the SERVERS VLAN port · static IP 192.168.20.10/24
2. Step 2 — Install dnsmasq and drop in /etc/dnsmasq.d/acme.conf with the Acme A records
3. Step 3 — Run the supplied Python http.server (port 8080) or install nginx / caddy
4. Step 4 — Update DHCP scope on the router for VLAN 10 to push DNS 192.168.20.10
5. Step 5 — Verify by name: nslookup, curl, ping server.acme.lan

Exercise 2 — Stage 5: Services + Wireless (2/2)

Optional Wi-Fi segment · deliverable · alternative tools

Optional Wi-Fi segment (60 min at home)

- Option 1 — GNS3 Cloud node bridged to the host's real Wi-Fi adapter · ★☆☆
- Option 2 — Mininet-WiFi software 802.11 emulator · ★★☆
- Option 3 — OpenWrt VM, full router / AP firmware · ★★★
- Bridge the Wi-Fi segment to a GUEST VLAN — internet-only, no SERVERS access

Deliverable

- Screenshot: curl http://www.acme.lan from a STAFF client
- Updated subnet table — now includes SERVERS + optional GUEST
- Optional: screenshot of Wi-Fi association

Alternative open-source tools

- DNS — Unbound · BIND9 · PowerDNS · Pi-hole
- HTTP — Caddy · Nginx · Lighttpd
- Wi-Fi — Mininet-WiFi · OpenWrt

References — Quick Index

Bookmark the authoritative sources

Standards & RFCs

- RFC 9293 — TCP (current revision)
- RFC 768 — UDP
- RFC 1035 — DNS
- RFC 9110 — HTTP semantics
- IEEE 802.11 family
- IEEE 802.1Q — VLAN tagging
- IANA service / port registry

Organisations & Tools

- Orgs: IETF · IEEE · IANA · ICANN · Wi-Fi Alliance
- Capture: Wireshark · tshark · tcpdump
- Simulation: GNS3 · VirtualBox · Mininet-WiFi
- DNS: dnsmasq · Unbound · BIND9 · Pi-hole
- HTTP: Caddy · Nginx · Lighttpd
- Router firmware: OpenWrt · OPNsense · pfSense
- Public DNS: Quad9 · Cloudflare 1.1.1.1 · Google 8.8.8.8

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End of Day 05

Application Services & Wi-Fi

Next session — Day 06 — Network Security & Course Wrap

From the Industry to the Classroom — Building the IT's Future, Today and Together!

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